MODULE 25 · DOMAIN 7 - TROUBLESHOOTING AND OPTIMISATION

Diagnosing Poor Outputs

The diagnostic sequence, symptoms and causes, one change at a time, the best baseline, plateaus, and fixes that are not the prompt.

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Diagnose before you change anything.

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The diagnostic sequence

01

Name it

The specific deficiency, against the criteria.

02

Compare

The output with what the prompt asked for.

03

Hypothesise

The most likely cause.

04

Change one thing

Only that cause.

05

Observe and record

Same input; compare; log it.

ON THE EXAM

Name and compare before you change. Record after you observe.

Name the deficiency precisely

- **Not:** "it's bad", "it's off", "it doesn't feel right".
- **Instead:** "answers two of the four questions", "uses contractions", "cites a policy that does not exist".
- **Measure against the criteria** from module six: accuracy, completeness, audience, format.
- **Point to where:** which section, which sentence, which figure.

> THE TIEBREAKER
>
> A deficiency you cannot name precisely, you cannot fix deliberately.

Symptoms, likely causes and first fixes

SYMPTOM	LIKELY CAUSE	FIRST FIX
Skips part of a multi-part ask	Questions not numbered	Number them; answer under each
Wrong tone	Tone undefined or contradictory	Voice rules and an example
Wrong format	Format never stated	State it; show an example
Invented facts	No source; no room for "I don't know"	Sources; allow it; quotes first
Stale information	Old file in knowledge	Update; remove the old version
Worse over many rounds	Drift from many changes	Revert to the best baseline

ON THE EXAM

Match the symptom to its cause before choosing the fix.

Compare the output with the prompt

- **Did the prompt actually ask for it?** Often the output faithfully follows a vague request.
- **Was the information provided?** Claude cannot use a contract it was never given.
- **Do the layers contradict?** Instructions, knowledge and the prompt pulling apart.
- **The colleague test:** would a capable new colleague, given only this prompt, have done better?

> THE TIEBREAKER
>
> Before blaming the output, read the prompt that produced it.

The unenumerated multi-part question

BEFORE

Four questions in one paragraph

- Cost, timeline, planning permission and approver, run together
- Claude answers cost and timeline
- Planning and approver silently missed
- The writer blames Claude

AFTER

Numbered, with a rule for gaps

- 1. Cost 2. Timeline 3. Planning 4. Approver
- "Answer each under its number"
- "If you cannot answer one, say so"
- All four answered, or a gap flagged

ON THE EXAM

Several questions in one paragraph: number them, and ask for each answer under its number.

One change at a time, with a log

- **Change one element per round:** audience, or format, or an example. Never all at once.
- **Hold everything else fixed:** same input, same model, same settings.
- **Record each round:** version, the change, the result against the criteria.
- **Keep what works,** and revert what does not.

> THE TIEBREAKER
>
> One change and a log make a good result repeatable. Five changes make it luck.

Quirk 15, settled: the prompt log

VERSION	CHANGE	RESULT AGAINST CRITERIA	KEEP?
v1	Recovered best earlier prompt	Baseline: misses exceptions	Baseline
v2	Named the audience: regional managers	Clearer, still misses exceptions	Keep
v3	Added "exceptions first" section	Exceptions now present	Keep
v4	Cut the length limit by half	Exceptions dropped again	Revert
v5	Numbered sections with a short example	Meets every criterion	Keep

THE SIGNAL

The log turns prompt roulette into a record anyone can repeat.

Drift and the best baseline

- **Drift:** after many rounds, quality slips below an earlier version.
- **Revert to the best baseline,** rather than patching the patches.
- **In a long chat,** drift can be context, so summarise and start fresh.
- **Keep versions:** without them, there is no baseline to return to.

> ON THE EXAM
>
> When the latest version is worse than an earlier one, go back. Do not keep digging.

Plateau: when tweaking stops helping

- **The signal:** several careful, single changes, and no improvement.
- **Stop rewording.** The wording is not the limit any more.
- **Change the approach:** decompose the task, add sources or examples, change the feature or the model.
- **Or change the split:** a person handles the part Claude cannot do well.

> THE TIEBREAKER
>
> A plateau means change approach, not more of the same.

When the fix is not the prompt

SYMPTOM	REAL CAUSE	FIX
Wrong totals in a workbook	Arithmetic done in Claude's head	Code execution
Cites an out-of-date policy	Old file in knowledge	Update; remove the old version
Shallow analysis of a complex case	Model tier too light	Step up a tier
Cannot answer about live bookings	Data not available	Provide the data, or escalate
Only one person's chats go wrong	Personal settings or styles	Check that person's layer
Contradictions late in a long chat	Context	Summarise and start fresh

ON THE EXAM

Some failures live in the feature, the model, the data or the settings. Look there too.

Diagnosis traps

THE TRAP	WHY IT FAILS
"It's bad" with no named deficiency	Nothing to aim the fix at
Rewriting everything at once	No idea which change mattered
Blaming Claude before reading the prompt	The prompt often asked for exactly this
Patching after drift sets in	Digs further from the best version
Rewording through a plateau	The wording is no longer the limit
A bigger model for a missing source	No model can use data it was not given

ON THE EXAM

Every trap treats before it diagnoses.

> Halden's North regional manager says the weekly service report from Claude is "just not good enough" and wants the prompt rewritten today.

What should happen first?

A Name the specific deficiency against the report's criteria: what is wrong, and where

B Rewrite the prompt from scratch

C Switch the Project to the most capable model

D Add "make it better" to the instructions

◆ PAUSE AND DECIDE

What should happen first?

A Name the specific deficiency against the report's criteria: what is wrong, and where
Diagnosis starts by naming the deficiency.

B Rewrite the prompt from scratch
Treats before diagnosing; prompt roulette.

C Switch the Project to the most capable model
A guess at the cause, with nothing named.

D Add "make it better" to the instructions
A vague instruction for a vague complaint.

THE SIGNAL

"Not good enough" is not a diagnosis: name the deficiency first.

> A client's facilities manager emailed four questions in one paragraph: cost, timeline, planning permission, and who should approve the work.
>
> Claude's drafted reply answers cost and timeline only.

What is the best fix?

A Ask Claude to try again

B Switch to a more capable model

C Number the four questions in the prompt, ask for an answer under each number, and say to flag any that cannot be answered

D Answer the missing questions by hand every time

◆ PAUSE AND DECIDE

What is the best fix?

A Ask Claude to try again

Same prompt, same risk.

B Switch to a more capable model

The cause is structure, not capability.

C Number the four questions in the prompt, ask for an answer under each number, and say to flag any that cannot be answered

Enumerate the parts, and make gaps visible.

D Answer the missing questions by hand every time

Permanent manual work for a fixable cause.

THE SIGNAL

Several questions run together: number them, and flag gaps.

> Halden's finance Project produces month-end summaries with two problems: some totals are wrong, and prices come from last year's rate card.
>
> Both the old and current rate cards are in the Project's knowledge.

Which two fixes address the real causes? Choose two.

A Ask Claude to be more careful with numbers

B Have code execution calculate the totals from the workbook

C Replace the old rate card in knowledge with the current one, and remove the old file

D Add "use the latest rates" to the prompt

E Switch the Project to a more capable model

◆ PAUSE AND DECIDE

Which two fixes address the real causes? Choose two.

A Ask Claude to be more careful with numbers

A plea, not a fix for in-head arithmetic.

B Have code execution calculate the totals from the workbook

Exact totals come from code.

C Replace the old rate card in knowledge with the current one, and remove the old file

Removes the stale source.

D Add "use the latest rates" to the prompt

The old file is still there to be retrieved.

E Switch the Project to a more capable model

Neither cause is about capability.

THE SIGNAL

Two symptoms, two non-prompt causes: code for totals, curation for stale prices.

Halden's bid manager sees that Claude's tender answers keep missing the client's mandatory compliance statements.

(1) Change one element of the prompt.

(2) Name the deficiency precisely.

(3) Form a hypothesis about the cause.

(4) Run the same tender questions again, compare, and record the result.

(5) Compare the output with what the prompt asked for.

What is the correct order?

A 5, 2, 3, 1, 4

B 2, 3, 5, 1, 4

C 2, 5, 1, 3, 4

D 2, 5, 3, 1, 4

◆ PAUSE AND DECIDE

What is the correct order?

A 5, 2, 3, 1, 4

Compares before knowing what to look for.

B 2, 3, 5, 1, 4

Hypothesises before checking the prompt.

C 2, 5, 1, 3, 4

Changes before forming a hypothesis.

D 2, 5, 3, 1, 4

Name, compare, hypothesise, change, observe.

THE SIGNAL

Name, compare, hypothesise, change one thing, observe and record.

> Quirk 15: for six weeks Halden's operations team has rewritten the weekly report prompt whenever a draft disappointed.
>
> Drafts are now worse than a version from three weeks ago, which someone remembers was good.

What is the best next step?

A Recover the best earlier version as the baseline, then change one element at a time and log each round

B Keep making new changes until the drafts improve

C Start again from a blank prompt

D Switch the Project to a different model

◆ PAUSE AND DECIDE

What is the best next step?

A Recover the best earlier version as the baseline, then change one element at a time and log each round

Revert to the best baseline; then controlled changes.

B Keep making new changes until the drafts improve

More roulette, further from the best version.

C Start again from a blank prompt

Throws away the best known version.

D Switch the Project to a different model

The method is the problem, not the model.

THE SIGNAL

Worse than an earlier version: go back to the best baseline, then change one thing at a time.

> Halden's bid team has made five careful, single changes to the prompt for complex technical tender questions, each logged.
>
> Quality has not improved for the last four rounds.

What is the best next move?

A Keep rewording the prompt with different phrases

B Add emphasis in capital letters to the key instructions

C Change approach: break each question into requirement, method and evidence, add the relevant method statement as a source, and consider a more capable model

D Stop using Claude for tenders altogether

◆ PAUSE AND DECIDE

What is the best next move?

A Keep rewording the prompt with different phrases

More of what has stopped working.

B Add emphasis in capital letters to the key instructions

Louder wording is still wording.

C Change approach: break each question into requirement, method and evidence, add the relevant method statement as a source, and consider a more capable model

A plateau calls for a different approach.

D Stop using Claude for tenders altogether

Gives up value that a better approach can unlock.

THE SIGNAL

Careful changes, no improvement: a plateau. Change the approach, not the wording.

> Customer success complains that Claude's replies ignore the response times in a client's contract.
>
> Comparing the output with the setup, the lead finds the contract was never added to the Project, and the prompt does not mention it.

What does this show, and what follows?

A Claude ignored the contract, so the instructions need stronger wording

B The model is too weak for contract work

C The instructions need key terms in capital letters

D The information was never provided; add the relevant contract terms to knowledge, then re-test

◆ PAUSE AND DECIDE

What does this show, and what follows?

A Claude ignored the contract, so the instructions need stronger wording

Claude cannot ignore what it never received.

B The model is too weak for contract work

No model can use data it was not given.

C The instructions need key terms in capital letters

Emphasis cannot supply missing information.

D The information was never provided; add the relevant contract terms to knowledge, then re-test

Missing input: provide it, then re-test.

THE SIGNAL

Compare with the prompt: the contract was never there. Provide it.

> Halden's HR policy Project gives full, well-structured answers to most supervisors.
>
> One manager always gets very short, bullet-only replies to the same kinds of question.

What is the most likely cause, and where should she look first?

A Random variation in the model; nothing can be done

B That manager's own settings: her chosen style or personal instructions; check them before changing the Project

C The Project instructions, which must be wrong for everyone

D Out-of-date knowledge in the Project

◆ PAUSE AND DECIDE

What is the most likely cause, and where should she look first?

A Random variation in the model; nothing can be done

A consistent pattern is not random.

B That manager's own settings: her chosen style or personal instructions; check them before changing the Project

Only one person affected: check that person's layer.

C The Project instructions, which must be wrong for everyone

Everyone else is fine, so the Project is not the cause.

D Out-of-date knowledge in the Project

Stale knowledge would affect everyone.

THE SIGNAL

Only one person's chats differ: look at that person's settings, not the Project.

In five lines

- **Diagnose first:** name it, compare with the prompt, hypothesise, change one thing, observe and record.
- **Match symptom to cause:** number multi-part questions; give sources; remove stale files.
- **One change at a time,** with a log; keep what works, revert what does not.
- **Drift:** revert to the best baseline. **Plateau:** change the approach.
- **Not every fix is the prompt:** code, knowledge, model, data, or one person's settings.

COMING UP

Next: feedback loops and workflow optimisation.